

Business Analytics Lab Record

IBM Cognos Analytics & Power BI – Practicals 1 to 10

Dataset: SuperStore (built from the Sample – Superstore sales data)

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About this record

This record works through all ten practicals in the lab manual using the 9,994-row Superstore sales file as the RetailPulse dataset. Every number, table and chart below was calculated directly from that file, so the figures can be cross-checked in Cognos Analytics and Power BI Desktop once the same steps are performed there. Places marked [Insert screenshot] are where your own screenshots from the tools belong.

Dataset and field mapping

The manual refers to a RetailPulse package with Region, Product Category, Store Name, Sales Amount and similar fields. The Superstore file has the same shape of data, so its columns are mapped as follows:

Lab manual field	Superstore column	Notes
Region	Region	4 values: Central, East, South, West (there is no "North")
Product Category	Category	Furniture, Office Supplies, Technology
Store Name / StoreID	State	Each of the 49 states is treated as one store/outlet
Sales Amount	Sales	US dollars
Sales Date	Order Date	3 Jan 2014 to 30 Dec 2017 (dates arrive as mixed text and are converted)
Quantity, OrderID, ProductID	Quantity, Order ID, Product ID	Used as is

Dataset facts	Value
Rows / columns	9,994 / 21
Distinct orders	5,009
Customers	793
Products	1,862
Stores (states)	49
Date range	3 Jan 2014 – 30 Dec 2017 (1,458 days)
Total Sales	\$2,297,200.86
Total Quantity	37,873
Total Profit	\$286,397.02

Adaptations made to the manual

- Region filter value "North" becomes "West" (Practicals 3 and 5) and the comparison region "South" is kept as is, because the data has no North region.
- "Current quarter" becomes Q4 2017 (1 Oct – 31 Dec 2017), the latest full quarter in the data.
- Row-level security roles (Practical 10) are North/South Manager in the manual; here they are West Manager and South Manager, with East and Central results also shown.
- Loyalty segments (Practical 6) are not in the data, so they are defined by number of distinct orders per customer: Bronze 1–4, Silver 5–7, Gold 8–10, Platinum 11 or more.
- The accompanying workbook RetailPulse_Sales.xlsx contains the Sales, Product, Store, Customer and Date tables that Practical 7 asks you to import.

Practical 1: Generate a First RetailPulse Report

Related topic: Introduction to IBM Cognos Analytics – Reporting

Objective

Navigate the Cognos Analytics Reporting environment and generate a running list report against the RetailPulse Sales package.

Materials required

IBM Cognos Analytics access, RetailPulse Sales package (loaded from RetailPulse_Sales.xlsx), a personal content folder.

Step-by-step procedure (applied to the RetailPulse data)

1. Open IBM Cognos Analytics and select New > Report.
2. Choose the RetailPulse Sales package as the data source.
3. Pick the List template.
4. Drag Region, Product Category and Sales Amount onto the canvas.
5. Click Run to preview live data.
6. Save as "RetailPulse – First Sales Report" in your personal folder.

Result

The list report returns one row per sales line: 9,994 rows, with Sales Amount summing to \$2,297,200.86. The first rows are:

Region	Product Category	Sales Amount
South	Furniture	\$261.96
South	Furniture	\$731.94
West	Office Supplies	\$14.62
South	Furniture	\$957.58
South	Office Supplies	\$22.37
West	Furniture	\$48.86
West	Office Supplies	\$7.28
West	Technology	\$907.15
West	Office Supplies	\$18.50
West	Office Supplies	\$114.90

Expected outcome check

- Three columns with correct labels and live rows (not placeholder text) – met: 9,994 rows.
- Report saved and reopens with the same live data connection.

Viva-voce answers

a) What is the main difference between Cognos Analytics Reporting and self-service dashboards?

Reporting produces precisely formatted, repeatable reports (often scheduled, printed or burst to many users) that a report author designs in detail. Dashboards are interactive visual canvases for exploration: widgets cross-filter each other and users can change what they see without editing the report.

b) Name the four tabs available on the Reporting side panel.

Source (the package contents), Data items (items already in the report), Toolbox (layout and object tools), and Properties (settings of the selected object). Exact tab names can differ slightly between Cognos versions, so confirm against your installed one.

c) Which authoring template would you choose for a pivot-style summary, and why?

The Crosstab template, because it puts one dimension on the rows, another on the columns and the measure in the cells, which is exactly a pivot layout.

d) What must you select before a new report can display any data?

A data source – a package (or data module) such as the RetailPulse Sales package.

e) Where in the interface would you rename or add a new report page?

In the Pages area (page explorer) of the report: add a page there and rename it from its properties or context menu.

Practical 2: Build a Grouped RetailPulse Sales List Report

Related topic: Create List Reports

Objective

Construct a list report, group it meaningfully, format its columns, and add a summarizing footer.

Materials required

IBM Cognos Analytics access, RetailPulse Sales package, report from Practical 1 (optional).

Step-by-step procedure (applied to the RetailPulse data)

1. Create a new List report on the RetailPulse Sales package.
2. Add Region, Store Name, Product Category, Quantity and Sales Amount.
3. Group by Region, then by Store Name.
4. Format Sales Amount as currency and right-align the numeric columns.
5. Add a list footer under Sales Amount using Total (sum).
6. Add a page header with the title and run date; run and save.

Result

Excerpt of the grouped report for the first two stores of the Central region. Repeated Region and Store values are suppressed by grouping, and each store ends with its total.

Region	Store Name	Product Category	Quantity	Sales Amount
Central	Illinois	Furniture	448	\$28,274.52
		Office Supplies	1,095	\$19,907.91
		Technology	302	\$31,983.67
	Total Illinois		1,845	\$80,166.10
	Indiana	Furniture	83	\$11,496.71
		Office Supplies	389	\$15,735.40
		Technology	106	\$26,323.25
		Total Indiana	578	\$53,555.36

Region-level totals and the grand total (the list footer):

Region	Quantity	Sales Amount
Central	8,780	\$501,239.89
East	10,618	\$678,781.24
South	6,209	\$391,721.91
West	12,266	\$725,457.82
Grand total	37,873	\$2,297,200.86

Expected outcome check

- Nested Region > Store bands, duplicates collapsed – met.
- Group totals sum correctly: the four regional totals add to the grand total of \$2,297,200.86.

Viva-voce answers

a) What happens to duplicate values in a column once it is grouped?

They are shown once per group (suppressed on later rows), so the repeated value acts as a group heading and the rows beneath it are its members.

b) Why would you group by Region before Store Name rather than the reverse?

A store belongs to one region, so Region is the broader level of the hierarchy. Grouping broad-to-narrow gives clean region bands with store sub-bands and meaningful subtotals at each level.

c) Which pane do you use to apply currency formatting to a column?

The Properties pane (Data Format property of the column), which opens the format dialog for currency, decimals and symbol.

d) Name two aggregate functions you could add to a list footer besides Total.

Average and Count (also Minimum or Maximum).

e) What is the practical difference between a page footer and a list footer?

A page footer repeats at the bottom of every printed page. A list footer appears once, directly under the list, and holds summaries of the data (for example the total of Sales Amount).

Practical 3: Filter RetailPulse Sales by Region and Date

Related topic: Focus Reports Using Filters

Objective

Apply a simple filter and an advanced detail filter to scope the report to one region and one period, then generalize it with a prompt.

Materials required

IBM Cognos Analytics access, the grouped list report from Practical 2.

Step-by-step procedure (applied to the RetailPulse data)

1. Open the grouped RetailPulse list report.
2. In the Filters pane add a simple filter: Region = "West".
3. Add an advanced detail filter: Sales Date between 2017-10-01 and 2017-12-31 (Q4 2017).
4. Combine both conditions with AND.
5. Run the report and confirm the row count drops.
6. Convert the Region filter into a prompted filter.

Filter expressions

```
[Region] = 'West'
[Sales Date] between 2017-10-01 and 2017-12-31    -- date literal syntax can vary by Cognos
version
Combined: [Region] = 'West' AND [Sales Date] between 2017-10-01 and 2017-12-31
```

Result

Scope	Rows	Sales Amount
Unfiltered report	9,994	\$2,297,200.86
Region = West only	3,203	\$725,457.82
Q4 2017 only (all regions)	1,219	–
West AND Q4 2017	394	\$79,806.32
Prompt re-run with Region = South (Q4 2017)	203	\$56,064.11

With both filters active, only 394 rows remain (quantity 1,574). Top stores in the filtered report:

Store Name	Sales Amount
California	\$48,675.28
Washington	\$17,010.54
Arizona	\$5,696.64
Colorado	\$4,174.99
Nevada	\$1,669.60

Changing the prompt value from West to South returns a different set of rows, which proves the prompted filter works for other users.

Viva-voce answers

a) What is the difference between a detail filter and a summary filter?

A detail filter removes individual rows before any aggregation happens. A summary filter is applied after aggregation, to grouped totals (for example only stores whose total sales exceed a value).

b) When would you choose an advanced detail filter over a simple condition?

When the condition needs an expression: ranges with BETWEEN, several data items, functions or calculations, or combinations of AND/OR that the simple filter dialog cannot express.

c) Write the logical operator you would use to require both conditions to be true.

AND.

d) Why might you convert a fixed filter value into a prompted filter?

So one report serves many users and scenarios: each person picks their own region or period at run time without the author editing the report.

e) What effect does filtering have on report performance for large RetailPulse datasets?

Detail filters are normally pushed to the database as a WHERE clause, so fewer rows are retrieved, transferred and rendered; the report runs faster and uses less memory.

Practical 4: Build a RetailPulse Region × Product Crosstab

Related topic: Create Crosstab Reports

Objective

Build a crosstab summarizing sales across Region and Product Category with two measures and summary totals.

Materials required

IBM Cognos Analytics access, RetailPulse Sales package.

Step-by-step procedure (applied to the RetailPulse data)

1. Start a new report with the Crosstab template.
2. Drag Region to the rows and Product Category to the columns.
3. Drag Sales Amount into the measure area.
4. Add Quantity as a second measure beside Sales Amount.
5. Turn on row and column summaries (grand totals).
6. Format Sales Amount as currency, run, and save as "RetailPulse Region-Product Crosstab".

Result

Region	Furniture Sales	Qty	Office Supplies Sales	Qty	Technology Sales	Qty	Total Sales	Qty
Central	\$163,797	1,827	\$167,026	5,409	\$170,416	1,544	\$501,240	8,780
East	\$208,291	2,214	\$205,516	6,462	\$264,974	1,942	\$678,781	10,618
South	\$117,299	1,291	\$125,651	3,800	\$148,772	1,118	\$391,722	6,209
West	\$252,613	2,696	\$220,853	7,235	\$251,992	2,335	\$725,458	12,266
Total	\$742,000	8,028	\$719,047	22,906	\$836,154	6,939	\$2,297,201	37,873

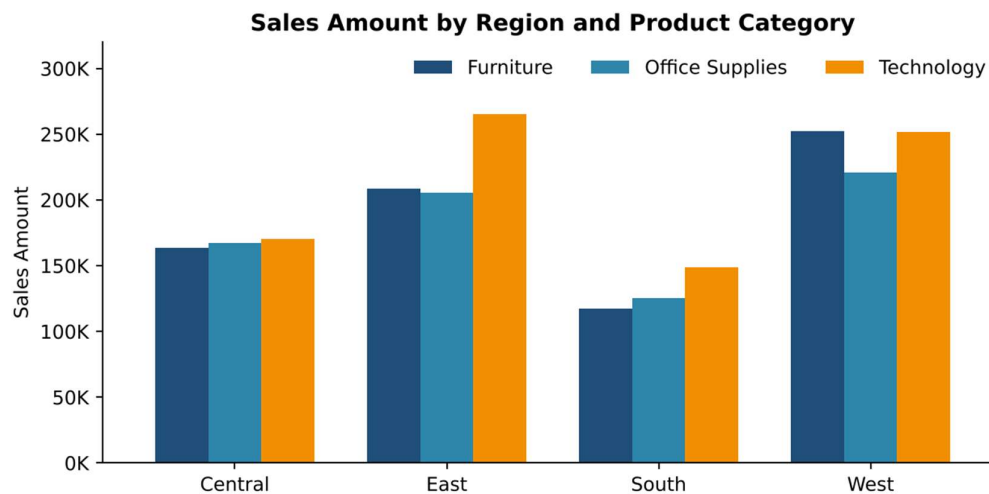


Figure 4.1 – Sales Amount by Region and Product Category

Reconciliation: the crosstab grand total (\$2,297,200.86 sales, 37,873 units) equals the grand total of the grouped list in Practical 2, as the manual requires.

Viva-voce answers

a) How does a crosstab report differ structurally from a list report?

A list shows each record as a row with fixed columns. A crosstab places one set of items on the rows, another on the columns, and shows the aggregated measure at each intersection, so columns grow with the data.

b) Which three drop zones define a basic crosstab layout?

Rows, Columns, and the Measure (cell values) area.

c) How would you add a second measure to an existing crosstab?

Drag the second measure (Quantity) into the measure area next to the first so the two nest side by side under each column.

d) Name two data source types that can power a Cognos crosstab report.

Relational data (packages/data modules over tables) and dimensional (OLAP) data sources.

e) What does turning on row and column summaries add to a crosstab?

Grand totals at the end of every row and column, so you can reconcile the table with overall totals.

Practical 5: Build the RetailPulse Sales Prompt Report

Related topic: Design a Cognos Prompt Report for Sales Analysis

Objective

Design a prompt page with value, date-range and cascading prompts, linked to report filters.

Materials required

IBM Cognos Analytics access, the crosstab from Practical 4.

Step-by-step procedure (applied to the RetailPulse data)

1. Open the Region-Product Crosstab report.
2. Insert a prompt page ahead of the crosstab page.
3. Add a Value prompt for Region and a Date-range prompt for the reporting period.
4. Add a cascading Store prompt that depends on the chosen Region.
5. Link each control to its filter on the crosstab page.
6. Set default values, add a Finish button, run and test.

Prompt page design

Control	Prompt type	Parameter	Filter on the crosstab page	Default
Region	Value (select)	?p_Region?	[Region] = ?p_Region?	West
Period	Date range	?p_Period?	[Sales Date] between ?p_Period?	2017-10-01 to 2017-12-31
Store	Value, cascades from Region	?p_Store?	[Store Name] in (?p_Store?)	All stores

Result: tested outputs

Region prompt	Furniture	Office Supplies	Technology	Total
West	\$32,119.31	\$27,162.52	\$20,524.49	\$79,806.32
South	\$19,678.20	\$15,292.79	\$21,093.12	\$56,064.11

Both runs use the date range 1 Oct – 31 Dec 2017. Choosing West returns \$79,806.32 and choosing South returns \$56,064.11, matching the filtered row totals from Practical 3.

Cascading store lists (what the Store prompt offers after each Region choice):

Region chosen	Stores offered	Examples
Central	13	Illinois, Indiana, Iowa, Kansas, ...
East	14	Connecticut, Delaware, District of Columbia, Maine, ...
South	11	Alabama, Arkansas, Florida, Georgia, ...
West	11	Arizona, California, Colorado, Idaho, ...

Viva-voce answers

a) What problem do prompts solve compared to a fixed, hard-coded filter?

A hard-coded filter has to be edited by the author for every new region or period. A prompt lets each viewer choose values at run time, so one report serves everybody.

b) Describe a scenario where a cascading prompt is preferable to two independent prompts.

Region and Store: once West is chosen, the Store list shows only the 11 western stores, so the user cannot pick an impossible combination such as West with Illinois.

c) What is the purpose of setting default values on a prompt control?

They give a sensible starting selection so the report can run without requiring input (for example in scheduled runs) and users see meaningful data immediately.

d) Which report element actually applies the viewer's prompted selections?

The filter on the report page whose expression references the prompt parameter, for example [Region] = ?p_Region?.

e) How would you test that a prompted report behaves correctly for multiple regions?

Run it once per region (and with defaults, multiple selections and an empty selection) and compare each output with the known totals from the unprompted report; the regional totals should add up to the overall total.

Practical 6: Build the RetailPulse Customer Insights Dashboard

Related topic: Build a Dashboard for Customer Insights Visualization

Objective

Assemble a dashboard with a summary card, trend chart, segment breakdown and a shared filter widget for customer data.

Materials required

IBM Cognos Analytics access, RetailPulse Customers data module (built from the Customer and Sales tables).

Step-by-step procedure (applied to the RetailPulse data)

1. Create a new Dashboard connected to the Customers data module.
2. Add a summary card for active customers.
3. Add a line chart of new vs returning customers over time.
4. Add a bar chart of customers by loyalty segment.
5. Add a Region filter widget and check it cross-filters every widget.
6. Arrange the widgets in a clear hierarchy and save.

Definitions used

- Active customers: distinct Customer ID with at least one order in the selected period.
- New customer in a year: the customer's first-ever order falls in that year; otherwise returning.
- Loyalty segment: Bronze 1–4 orders, Silver 5–7, Gold 8–10, Platinum 11+ (an assumption, as the data has no loyalty field).



Figure 6.1 – Dashboard layout and values calculated from the data

Year	Active customers	New	Returning
2014	595	595	0

Year	Active customers	New	Returning
2015	573	136	437
2016	638	51	587
2017	693	11	682

Over the whole period there are 793 customers, 98.5% of them ordered more than once, and revenue per customer is \$2,896.85. Customer counts by Region (629 Central, 674 East, 512 South, 686 West) add to more than 793 because a customer can buy in several regions.

Viva-voce answers

a) What is the key interactive behavior that distinguishes a dashboard from a static report?

Cross-filtering and interaction: selecting a value or filter updates every widget live, and users can drill, slice and explore without a new run.

b) Which widget type would you use to show one prominent number, such as active customers?

A summary (KPI) card.

c) Explain how a shared filter widget affects the other widgets on a dashboard.

The selection becomes a filter applied to all widgets that use the same data, so choosing Region = West makes the card, trend and bar chart all show only West customers.

d) Name two customer-focused KPIs suitable for a RetailPulse insights dashboard.

Active customers and repeat-customer rate (others: revenue per customer, retention rate, customer lifetime value).

e) Why might a trend chart reveal something a single summary card cannot?

The card is one snapshot. The trend shows direction and change: here new customers fall from 595 to 11 a year while returning customers rise, which a single total would hide.

Practical 7: Import and Transform RetailPulse Data in Power BI

Related topic: Power BI Interface, Data Import and Transformation (Unit 3)

Objective

Explore Power BI Desktop, import the RetailPulse workbook and clean and shape it in Power Query.

Materials required

Power BI Desktop (free), RetailPulse_Sales.xlsx with Sales, Product and Store tables (supplied).

Step-by-step procedure (applied to the RetailPulse data)

1. Open Power BI Desktop and identify Report, Table and Model views, the ribbon, and the Data, Visualizations and Filters panes.
2. Home > Get data > Excel workbook, tick Sales, Product and Store, click Transform Data.
3. Set data types: Sales Date as Date, Quantity as Whole Number, Sales Amount as Fixed decimal number.
4. Remove blank and duplicate rows from Sales, replace null Quantity with 0, rename unclear headings.
5. Add a Conditional Column "Sale Size": "Large" when Sales Amount ≥ 1000 , otherwise "Standard".
6. Review Applied Steps, Close & Apply, save as RetailPulse_Analysis.pbix.

Data profiling results

Table	Rows	Columns	Key columns
Sales	9,994	13	OrderID, ProductID, StoreID, Sales Date, Quantity, Sales Amount, Profit
Product	1,862	4	ProductID, Product Category, Sub-Category, Product Name
Store	49	4	StoreID, Store Name, Region, Country

Check	Result
Null values in the source data	0 – so replacing null Quantity with 0 changes nothing, but the step is still recorded
Completely blank rows	0
Exact duplicate rows (all columns)	0, because every Row ID is unique
Same row ignoring Row ID	1 pair – a repeated line item, kept because it may be a genuine second purchase
Date text format	Mixed (e.g. 11-08-2016 and 4/15/2017); converted to proper dates before typing
Quantity data type	Whole number
Sales Amount data type	Fixed decimal number

Sale Size column

Sale Size	Rows	Sales Amount
Large (Sales Amount ≥ 1000)	468	\$990,169.08
Standard	9,526	\$1,307,031.78

Applied Steps (Sales query)

1. Source
2. Navigation
3. Promoted Headers
4. Changed Type
5. Removed Blank Rows

6. Removed Duplicates
7. Replaced Value (null → 0 in Quantity)
8. Added Conditional Column (Sale Size)

The full Power Query M script is in Appendix A. The query is saved so it re-runs automatically on refresh.

Viva-voce answers

a) What is the difference between Import mode and DirectQuery mode in Power BI?

Import copies the data into the report's in-memory model: fast and fully featured, but it needs a refresh to update. DirectQuery leaves data at the source and sends queries at run time: always current, but slower and with fewer modelling options.

b) What are Applied Steps in Power Query, and why are they useful?

An ordered record of every transformation. They can be reviewed, edited, reordered or deleted, and are replayed automatically on each data refresh, so cleaning is repeatable.

c) Which of the three views would you use to check relationships between tables?

Model view.

d) Why must data types be set correctly before building visuals or calculations?

Types decide which operations, sorting, aggregations and date hierarchies are available. Text-typed dates or amounts cause wrong sorting, failed sums and broken time calculations.

e) What is the difference between the Transform Data and Close & Apply commands?

Transform Data opens Power Query Editor so you can shape the data. Close & Apply saves those transformations and loads the result into the data model.

Practical 8: Model, Calculate, and Publish a RetailPulse Sales Report

Related topic: Relationships, Basic DAX, Visualizations, and Publishing Reports (Unit 3)

Objective

Create relationships, write basic DAX measures and a calculated column, build interactive visuals and publish to the Power BI service.

Materials required

Power BI Desktop, RetailPulse_Analysis.pbix from Practical 7, a Power BI service account.

Step-by-step procedure (applied to the RetailPulse data)

1. In Model view create one-to-many relationships Product[ProductID] → Sales[ProductID] and Store[StoreID] → Sales[StoreID], single-direction filtering.
2. Create measures Total Sales, Total Quantity and Average Order Value.
3. Create the calculated column Sales per Unit in the Sales table.
4. Build a page: card for Total Sales, bar chart by Product Category, line chart by month, Region slicer.
5. Click a bar and use the slicer to confirm cross-filtering; add titles and axis labels.
6. Home > Publish, choose a workspace, open the report in the service and share it, subject to permissions.

Model

From (one side)	To (many side)	Cardinality	Filter direction
Product[ProductID]	Sales[ProductID]	One-to-many (1:*)	Single
Store[StoreID]	Sales[StoreID]	One-to-many (1:*)	Single

DAX

```
Total Sales = SUM(Sales[Sales Amount])
Total Quantity = SUM(Sales[Quantity])
Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
Sales per Unit = DIVIDE(Sales[Sales Amount], Sales[Quantity]) -- calculated column
```

Result

Measure	Value with no slicer selection
Total Sales	\$2,297,200.86
Total Quantity	37,873
Distinct orders	5,009
Average Order Value	\$458.61

Sales per Unit for the first rows of the Sales table:

OrderID	Sales Amount	Quantity	Sales per Unit
CA-2016-152156	\$261.96	2	\$130.98
CA-2016-152156	\$731.94	3	\$243.98
CA-2016-138688	\$14.62	2	\$7.31
US-2015-108966	\$957.58	5	\$191.52
US-2015-108966	\$22.37	2	\$11.18
CA-2014-115812	\$48.86	7	\$6.98

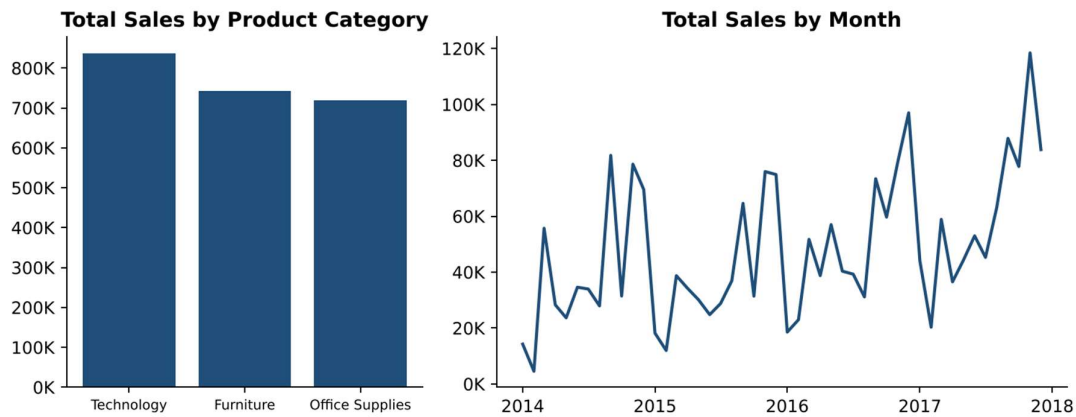


Figure 8.1 – Total Sales by Product Category and by month

Product Category	Total Sales
Technology	\$836,154.03
Furniture	\$741,999.80
Office Supplies	\$719,047.03
Total	\$2,297,200.86

Region (slicer value)	Total Sales card shows
West	\$725,457.82
East	\$678,781.24
Central	\$501,239.89
South	\$391,721.91

Check: the Total Sales card equals the sum of the three category bars (\$2,297,200.86), and the card changes to the regional values above when a Region slicer is selected.

Viva-voce answers

a) Why is a one-to-many relationship from a dimension table to a fact table preferred?

Each dimension row is unique, so one Store or Product row filters many Sales rows without ambiguity. This is the star-schema pattern: less duplicated data, faster queries and correct filter propagation.

b) What is the difference between a measure and a calculated column?

A calculated column is computed row by row when data is refreshed and stored in the table. A measure is computed at query time from the current filter context and is not stored, so it responds to slicers and visuals.

c) Why is the DIVIDE function used instead of the / operator?

DIVIDE handles a zero or blank denominator safely, returning blank (or a chosen alternate value) instead of an error or infinity.

d) Which visual would you choose to show a trend over time, and why?

A line chart, because the connected line shows direction, seasonality and turning points across ordered time periods.

e) What must be in place before another person can view your published report?

They need access to the workspace or report (shared with them) and a suitable licence – Pro or Premium Per User, unless the workspace is on Premium capacity.

Practical 9: Time Intelligence and Advanced DAX for RetailPulse

Related topic: Advanced Data Modeling, Time Intelligence, Complex DAX Functions, and Performance Optimization (Unit 4)

Objective

Create a date table, build time-intelligence and context-modifying DAX measures, and use Performance Analyzer to find and reduce slow visuals.

Materials required

Power BI Desktop, RetailPulse_Analysis.pbix from Practical 8. The data covers four calendar years (2014–2017), enough for year-on-year measures.

Step-by-step procedure (applied to the RetailPulse data)

1. Create the Date table with CALENDAR, add Year, Month Number and Month Name, mark it as a date table and relate Date[Date] to Sales[Sales Date].
2. Create Sales YTD and Sales LY.
3. Create YoY Growth % with VAR and RETURN.
4. Create Region Share % with CALCULATE and ALL.
5. Build a matrix with Year and Month on rows and the four measures as values.
6. Record with View > Performance Analyzer, optimize, and record again.

DAX

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))
Year = YEAR('Date'[Date])
Month Number = MONTH('Date'[Date])
Month Name = FORMAT('Date'[Date], "MMMM")

Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % =
VAR Curr = [Total Sales]
VAR Prev = [Sales LY]
RETURN DIVIDE(Curr - Prev, Prev)
Region Share % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))
```

The Date table runs from 3 Jan 2014 to 30 Dec 2017 and has 1,458 rows (the Date sheet in the workbook shows the same range).

Result: yearly and monthly values

Year	Total Sales	Sales LY	YoY Growth %
2014	\$484,247.50	blank	blank
2015	\$470,532.51	\$484,247.50	-2.8%
2016	\$609,205.60	\$470,532.51	29.5%
2017	\$733,215.26	\$609,205.60	20.4%

Matrix for 2017 (Sales YTD restarts in January, and Sales LY is the same month of 2016):

Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
January	\$43,971.37	\$43,971.37	\$18,542.49	137.1%
February	\$20,301.13	\$64,272.51	\$22,978.82	-11.7%
March	\$58,872.35	\$123,144.86	\$51,715.88	13.8%
April	\$36,521.54	\$159,666.40	\$38,750.04	-5.8%
May	\$44,261.11	\$203,927.51	\$56,987.73	-22.3%

Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
June	\$52,981.73	\$256,909.23	\$40,344.53	31.3%
July	\$45,264.42	\$302,173.65	\$39,261.96	15.3%
August	\$63,120.89	\$365,294.54	\$31,115.37	102.9%
September	\$87,866.65	\$453,161.19	\$73,410.02	19.7%
October	\$77,776.92	\$530,938.11	\$59,687.75	30.3%
November	\$118,447.82	\$649,385.94	\$79,411.97	49.2%
December	\$83,829.32	\$733,215.26	\$96,999.04	-13.6%

For 2014 the Sales LY and YoY Growth % cells are blank because there is no 2013 data, as the manual expects.

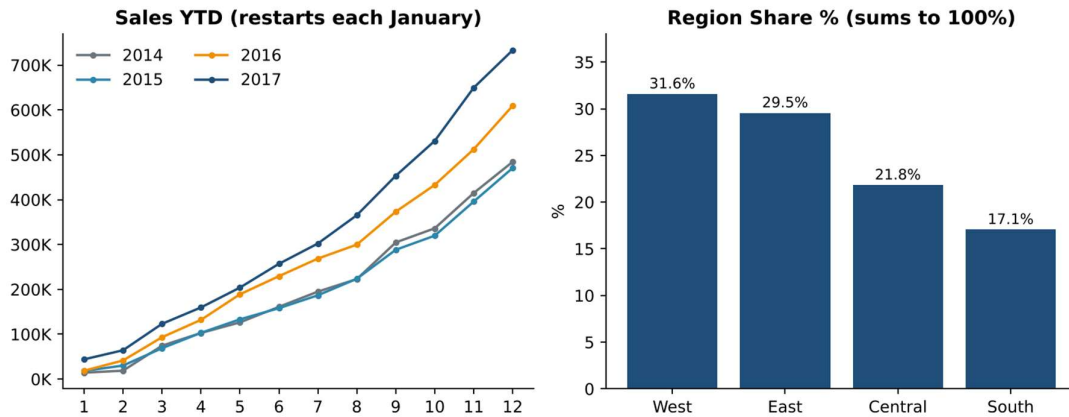


Figure 9.1 – Sales YTD by year and Region Share %

Region	Region Share %
West	31.6%
East	29.5%
Central	21.8%
South	17.1%
Total	100.0%

Performance Analyzer

Performance Analyzer timings depend on the machine, so they have to be recorded in your own Power BI Desktop session; no timings are given here. Record the first run, apply the changes below, then record again and fill the table.

Visual	Duration before (ms)	Duration after (ms)
Matrix		
Region Share % visual		
Cards / other visuals		

- Remove columns not used by any visual or measure (for example Ship Date, Product Name if unused).
- Reduce the number of visuals on the page, or split into pages.
- Keep the model a star schema with single-direction relationships.
- Use measures instead of calculated columns where possible.

Viva-voice answers

a) Why does time intelligence require a separate, continuous date table?

Functions like TOTALYTD and SAMEPERIODLASTYEAR move along a date column and need every date present with no gaps, plus a table marked as a date table. Sales dates alone have missing days, which would break the shifting.

b) What does CALCULATE do, and how does it change filter context?

CALCULATE evaluates an expression in a modified filter context: its filter arguments add to, replace or remove the filters coming from the report (for example SAMEPERIODLASTYEAR shifts the date filter back a year).

c) What is the benefit of using variables (VAR) in a DAX measure?

Readability and efficiency: a value is calculated once, named, and reused, which also makes the formula easier to debug.

d) What does the ALL function remove, and why is it used in Region Share %?

ALL removes filters from the given table or columns. In the denominator it removes the Region filter so every region is divided by the total of all regions, which makes the shares add to 100%.

e) Name two ways to improve the performance of a slow Power BI report.

Remove unused columns and reduce high-cardinality columns, and cut the number of visuals per page.
Other options: star schema, measures instead of calculated columns, Import mode.

Practical 10: Interactive Dashboard, Row-Level Security, and Power Platform Integration

Related topic: Interactive Dashboards, Row-Level Security, Power Apps, and Power Automate (Unit 4)

Objective

Build an interactive report with drill-through and bookmarks, restrict data by region with Row-Level Security, and connect the report to Power Automate and Power Apps.

Materials required

Power BI Desktop, the file from Practical 9, a service workspace with Pro or PPU licence, Power Automate and Power Apps access (work or school account), two test accounts. If licences are unavailable the instructor may demonstrate steps 5 and 6.

Step-by-step procedure (applied to the RetailPulse data)

1. Add Year and Region slicers, a drill-through page "Store Detail" (drill-through field Store Name), and two bookmarks with buttons to switch between a chart view and a table view.
2. Modeling > Manage roles: create West Manager and South Manager with table filters on Store.
3. Use View as to test each role.
4. Publish, assign test accounts to roles in the dataset Security settings, pin the Total Sales card to a dashboard.
5. Set a data alert on the pinned card and create a Power Automate flow with the trigger "When a data driven alert is triggered" to send an email or Teams message.
6. Add a Power Apps visual to Store Detail, pass Store Name and Sales Amount, choose Create new.

Role definitions

West Manager:	[Region] = "West"	-- table filter on Store
South Manager:	[Region] = "South"	-- table filter on Store

Result: what View as should show

Role / region	Stores visible	Sales rows	Total Sales shown
(extra check) Central	13	2,323	\$501,239.89
(extra check) East	14	2,848	\$678,781.24
South Manager	11	1,620	\$391,721.91
West Manager	11	3,203	\$725,457.82
No role (full access)	49	9,994	\$2,297,200.86

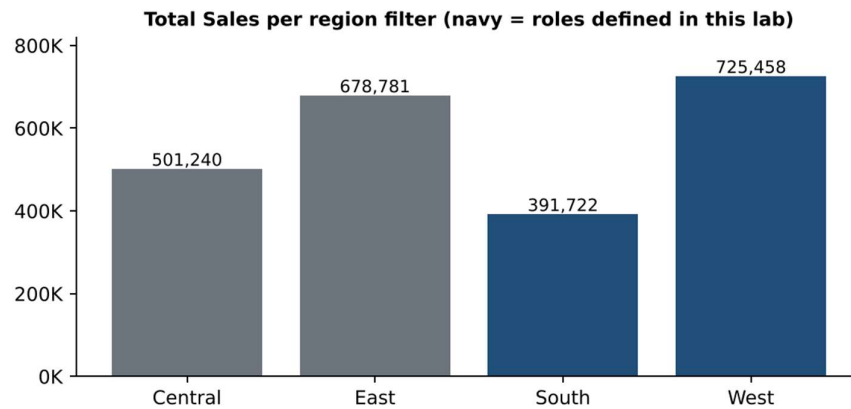


Figure 10.1 – Total Sales under each region filter (East and Central shown only as extra checks)

Drill-through example: right-clicking California in a visual and choosing Drill through > Store Detail opens the page filtered to that store. For California the Store Detail totals are Furniture \$156,064.60, Office Supplies \$142,351.95 and Technology \$159,271.08, which add to \$457,687.63.

Power Automate and Power Apps

- Alert: on the pinned Total Sales card choose Manage alerts and set a threshold; the alert fires when the value crosses it.
- Flow: in Power Automate use the trigger "When a data driven alert is triggered", select the alert, and add an action to send an email or post a Teams message.
- Power Apps visual: add it to Store Detail, give it the Store Name and Sales Amount fields, select Create new, and the generated app shows the selected store's details.

These last two parts need licensed accounts and live cloud services, so they cannot be reproduced from the data alone; run them in your own environment or ask the instructor to demonstrate them.

Viva-voice answers

a) What is Row-Level Security, and how are roles defined in Power BI?

RLS limits which rows each user can see. Roles are defined in Desktop under Modeling > Manage roles using DAX table-filter expressions (for example `[Region] = "West"`), then users or groups are assigned to roles in the Power BI service.

b) Why is a user with Contributor or Admin access to the workspace not restricted by RLS?

Those roles can edit the dataset, so Power BI does not apply RLS to them. RLS is enforced only for users with Viewer access (or read access through sharing).

c) How does drill-through differ from drill-down?

Drill-down moves to the next level of a hierarchy inside the same visual (year to quarter to month). Drill-through jumps to a different, detailed page filtered by the selected item, such as one store.

d) What does a Power Automate flow add to a Power BI data alert?

A plain alert only notifies inside Power BI (and by email). A flow turns the alert into an automated process: it can send Teams messages, create tasks, update records or start approvals.

e) Describe one business use for a Power Apps visual inside a Power BI report.

A store manager sees a store's sales on the report and, without leaving it, logs a stock request or a follow-up action for that store.

Appendix A: Power Query M script (Practical 7)

```
let
    Source = Excel.Workbook(File.Contents("C:\Path\To\RetailPulse_Sales.xlsx"), null, true),
    SalesSheet = Source{[Item="Sales", Kind="Sheet"]}[Data],
    Promoted = Table.PromoteHeaders(SalesSheet, [PromoteAllScalars=true]),
    Changed = Table.TransformColumnTypes(Promoted, {
        {"Sales Date", type date}, {"Ship Date", type date},
        {"Quantity", Int64.Type}, {"Sales Amount", Currency.Type},
        {"Discount", type number}, {"Profit", type number}}),
    NoBlankRows = Table.SelectRows(Changed, each not
List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null}))),
    NoDuplicates = Table.Distinct(NoBlankRows),
    NoNullQty = Table.ReplaceValue(NoDuplicates, null, 0, Replacer.ReplaceValue, {"Quantity"}),
    SaleSize = Table.AddColumn(NoNullQty, "Sale Size", each if [Sales Amount] >= 1000 then "Large"
else "Standard", type text)
in
    SaleSize
```

Appendix B: Notes and assumptions

- All amounts are in US dollars, as in the source file.
- Row counts and totals use all 9,994 rows; nothing was removed during cleaning because the data has no nulls, blanks or exact duplicates.
- Each state is treated as a store, so "Store Name" values are state names.
- Loyalty segments are defined by order count (Practical 6) because the data has no loyalty field.
- Cognos and Power BI screenshots, the saved Cognos reports, the .pbix file, Performance Analyzer timings and the Power Automate / Power Apps parts must be done in those tools; this record provides the data, expected values and steps for each.
- Cognos menu and tab names can differ slightly between versions.